

Technical University of Denmark

Advanced machine learning (02460)

Practice Exam Set B

Exam duration: 2 hours. Aid: None.

IMPORTANT: Questions are self-contained; all necessary formulas are provided.

Only information written on pages 2–11 will be evaluated.

Write your student number on all pages at the bottom-left.

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Part I Generative models

Exercise A — Variational autoencoder (VAE) with MoG prior

Consider a deep latent variable model with $z \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^2$. Let z follow a *Mixture of Gaussians* (MoG) prior

$$p(z) = \frac{1}{2} \mathcal{N}(z \mid 0, 1) + \frac{1}{2} \mathcal{N}(z \mid 4, 1),$$

and the likelihood be

$$p(\mathbf{x} \mid z) = \mathcal{N}(\mathbf{x} \mid [z, -z]^\top, \mathbf{I}_2).$$

Consider the variational posterior

$$q(z \mid \mathbf{x}) = \mathcal{N}(z \mid x_1 - \frac{x_2}{2}, 1).$$

Recall that the one-sample Monte Carlo estimate of the ELBO is

$$\text{ELBO}(\mathbf{x}) \approx \ln p(\mathbf{x} \mid z') + \ln p(z') - \ln q(z' \mid \mathbf{x}), \quad (1)$$

that the univariate Gaussian density is $\mathcal{N}(z \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(z-\mu)^2}{2\sigma^2})$, and the bivariate Gaussian density with identity covariance is $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{I}_2) = \frac{1}{2\pi} \exp(-\frac{1}{2}\|\mathbf{x} - \boldsymbol{\mu}\|^2)$.

Question A.1: With sampled value $z' = 2$ and $\mathbf{x}' = [2, -2]^\top$, what is the one-sample ELBO estimate?

- A ☐ $-\ln(2\pi) - \frac{3}{2}$
- B ☐ $-\ln(2\pi) - \frac{5}{2}$
- C ☐ $-2\ln(2\pi) - \frac{3}{2}$
- D ☐ $-\ln(2\pi) - \frac{1}{2}$

Give a brief argument for your answer.

Exercise B — Normalizing flows (affine coupling)

Consider the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{u}) = \begin{bmatrix} u_1, & e^{s(u_1)}u_2 + t(u_1) \end{bmatrix}^\top, \quad (2)$$

where $s, t : \mathbb{R} \rightarrow \mathbb{R}$ are continuously differentiable (C^1) functions. The base distribution is $p_{\mathbf{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid \mathbf{0}, \mathbf{I}_2)$.

Question B.1: Which of the following statements correctly describes T ?

- A ☐ T is invertible only if s is a constant function.
- B ☐ T is invertible only if $e^{s(u_1)} \neq 0$ for all u_1 .
- C ☐ T is invertible for *any* C^1 functions s, t .
- D ☐ T is not invertible in general.

Give a brief justification (derive T^{-1} or show it fails).

Question B.2: What is $\det J_T(\mathbf{u})$?

- A ☐ $\det J_T = 1$
- B ☐ $\det J_T = e^{s(u_1)}$
- C ☐ $\det J_T = e^{s(u_1)} + t(u_1)$
- D ☐ $\det J_T = e^{s(u_1)+t(u_1)}$

Give a brief justification.

Question B.3: Let $s(u_1) = 0$ and $t(u_1) = u_1^2$. What is $p_{\mathbf{x}}([1, 2]^\top)$?

- A ☐ $\frac{e^{-1}}{2\pi}$
- B ☐ $\frac{e^{-1/2}}{2\pi}$
- C ☐ $\frac{e^{-2}}{2\pi}$

$$\mathbf{D} \propto \frac{1}{2\pi}$$

Show the main steps of your derivation.

Exercise C — Denoising Diffusion Probabilistic Models

Consider a DDPM with $T = 5$ steps where $\mathbf{x}_0 \in \mathbb{R}^D$ is the data. Recall the forward process

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \quad q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}). \quad (3)$$

Question C.1: Which one of the following equalities is true for *any* variance schedule $\{\beta_t\}$?

- A ☐ $q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_1, \mathbf{x}_5) = q(\mathbf{x}_3 \mid \mathbf{x}_1)$
- B ☐ $q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_1, \mathbf{x}_5) = q(\mathbf{x}_3 \mid \mathbf{x}_5)$
- C ☐ $q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_1, \mathbf{x}_5) = q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_1)$
- D ☐ $q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_1, \mathbf{x}_5) = q(\mathbf{x}_3 \mid \mathbf{x}_4, \mathbf{x}_5)$

Give a brief justification using the Markov property of the forward process.

Part II Representation learning

Exercise D — Pull-back metrics

Consider the generative model $f : \mathbb{R}^d \rightarrow \mathbb{R}^D$ ($D \geq d$)

$$f(\mathbf{x}) = W_2 \tanh(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2, \quad (4)$$

where $W_1 \in \mathbb{R}^{D \times d}$, $W_2 \in \mathbb{R}^{D \times D}$ are full-rank matrices and $\tanh$ is applied element-wise. Recall that $\frac{d}{dx} \tanh(x) = 1 - \tanh^2(x)$. Define

$$\Lambda = \text{diag}(\mathbf{1} - \tanh^2(W_1 \mathbf{x} + \mathbf{b}_1)) \in \mathbb{R}^{D \times D}. \quad (5)$$

Question D.1: The pull-back metric $G_{\mathbf{x}} = J_{\mathbf{x}}^\top J_{\mathbf{x}}$ (where $J_{\mathbf{x}}$ is the Jacobian of f) is:

- A ☐ $G_{\mathbf{x}} = W_1^\top \Lambda W_2^\top W_2 \Lambda W_1$
- B ☐ $G_{\mathbf{x}} = W_2^\top \Lambda W_1^\top W_1 \Lambda W_2$
- C ☐ $G_{\mathbf{x}} = W_1^\top \Lambda^2 W_2^\top W_2 W_1$
- D ☐ $G_{\mathbf{x}} = W_1^\top W_2^\top W_2 \Lambda^2 W_1$

Give a brief argument for your answer.

Exercise E — Curve energy

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \begin{bmatrix} 2x_1^2, & x_1^2 + 3x_2^2, & 2x_1 + x_2^2 \end{bmatrix}^\top. \quad (6)$$

Consider also the two latent curves $\mathbf{c}_1(t) = t[1, 0]^\top$ and $\mathbf{c}_2(t) = t[0, 1]^\top$ for $t \in [0, 1]$.

Question E.1: Derive the pull-back metric $G_{\mathbf{x}} = J_{\mathbf{x}}^\top J_{\mathbf{x}}$.

Question E.2: Evaluate the energies $E(\mathbf{c}_1)$ and $E(\mathbf{c}_2)$ using

$$E(\mathbf{c}) = \int_0^1 \dot{\mathbf{c}}(t)^\top G_{\mathbf{c}(t)} \dot{\mathbf{c}}(t) \, dt. \quad (7)$$

$$E(\mathbf{c}_1) =$$

$$E(\mathbf{c}_2) =$$

Sketch your derivations below.

Exercise F — Bregman divergence

In information geometry, the Bregman divergence for comparing distributions p and q is

$$D_F(p, q) = F(p) - F(q) - \langle \nabla F(q), p - q \rangle, \quad (8)$$

where $\nabla F(q) = [\partial F / \partial q_1, \dots, \partial F / \partial q_D]^\top$.

Consider two probability vectors $p = [p_1, \dots, p_D]^\top$ and $q = [q_1, \dots, q_D]^\top$ with $p_i > 0$, $q_i > 0$, and $\sum_i p_i = \sum_i q_i = 1$.

Question F.1: Let $F(p) = \sum_i p_i \ln p_i$ (negative entropy). Show that $D_F(p, q) = \text{KL}(p \| q)$, where $\text{KL}(p \| q) = \sum_i p_i \ln(p_i / q_i)$.

Hint: First compute $\nabla F(q)$.

Part III Graph models

Exercise G — Graph metrics

Consider the *path graph* with 5 nodes $a-b-c-d-e$.

The *clustering coefficient* of a node u is defined as

$$c_u = \frac{|\{(v_1, v_2) \in E : v_1, v_2 \in \mathcal{N}(u)\}|}{\binom{d_u}{2}}, \quad (9)$$

where d_u is the degree of u (use $c_u = 0$ by convention when $d_u < 2$).

Question G.1: What are the clustering coefficients of node b and node a ?

$$c_b = \qquad \qquad \qquad c_a =$$

Show the main steps.

The *eigenvector centrality* satisfies $e_u = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} e_v$, where λ is the largest eigenvalue of the adjacency matrix. For the path graph above, $e_a = e_e = 0.29$ and $e_b = e_d = 0.50$.

Question G.2: What is the eigenvector centrality e_c ?

$$e_c =$$

Show the main steps.

Exercise H — Graph Fourier transform and convolution

Consider a *path graph* with three nodes: a – b – c . The adjacency matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Its eigendecomposition is $A = U\Lambda U^\top$ with

$$\Lambda = \text{diag}(-\sqrt{2}, 0, \sqrt{2}), \quad U = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{2} & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{bmatrix}.$$

A graph convolution with filter $h[k]$ can be written as

$$\mathbf{y} = \sum_{k=0}^N h[k] A^k \mathbf{x} = U \left(\sum_{k=0}^N h[k] \Lambda^k \right) U^\top \mathbf{x}. \quad (10)$$

Question H.1: Use the filter $h[k] = 1$ for $0 \leq k \leq 2$ and $h[k] = 0$ otherwise. Given vertex-domain signal $\mathbf{x} = [1, 0, 1]^\top$, compute the filtered signal $\mathbf{y}$.

$$\mathbf{y} = \begin{bmatrix} & & \end{bmatrix}^\top$$

Show the main steps.

Question H.2: For the same filter, what is the Fourier-domain filter matrix $M = \sum_{k=0}^N h[k] \Lambda^k$ (a 3×3 diagonal matrix)?

$$M = \text{diag}(, ,)$$

Show the main steps.

Exercise I — Weisfeiler-Lehman test

Consider a graph with 5 nodes: a 4-cycle $a-b-c-d-a$ with an additional edge $b-d$ (making a cycle with a chord). Run the WL algorithm with:

1. Initial labels $\ell_v^{(0)} = d_v$ (degree).
2. Update: $\ell_v^{(i)} = \text{hash}(\{\{\ell_u^{(i-1)} : u \in \mathcal{N}(v)\}\})$.

Question I.1: Which of the following are true? Select all that apply.

- A** ☐ $\ell_a^{(0)} = \ell_c^{(0)} = 2$
B ☐ $\ell_b^{(0)} = \ell_d^{(0)} = 3$
C ☐ $\ell_a^{(1)} = \ell_c^{(1)}$
D ☐ $\ell_b^{(1)} = \ell_d^{(1)}$
E ☐ $\ell_a^{(1)} = \ell_b^{(1)}$

Question I.2: After how many rounds does the WL algorithm halt?

Number of rounds =

Give a brief argument.

Exercise J — Message passing

Consider a 4-cycle $a-b-c-d-a$. We use the GNN

$$h_u^{(k)} = w_0 + w_1 h_u^{(k-1)} + w_2 \sum_{v \in \mathcal{N}(u)} h_v^{(k-1)}, \quad (11)$$

with $w_0 = 0$, $w_1 = 1$, $w_2 = 1$, and initial features $h_a^{(0)} = 1$, $h_b^{(0)} = 2$, $h_c^{(0)} = 1$, $h_d^{(0)} = 2$.

Question J.1: What are the node features after one message-passing round?

$$h_a^{(1)} = \quad h_b^{(1)} = \quad h_c^{(1)} = \quad h_d^{(1)} =$$

Show the main steps.

Question J.2: Write the single-layer update as $\mathbf{h}^{(k)} = M \mathbf{h}^{(k-1)}$ (with $w_0 = 0$). Fill in

all non-zero entries of M .

$$M = \begin{bmatrix} & \\ & \end{bmatrix}$$

Question J.3: Choose parameters w_0, w_1, w_2 such that the GNN implements the identity, i.e. $h_u^{(1)} = h_u^{(0)}$ for all u .

$$w_0 = \quad w_1 = \quad w_2 =$$